FinMini: Two-Stage Fine-Tuning of a Compact LLM for Financial Sentiment Analysis and Text-Driven Stock Movement Prediction

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Abstract

Large language models (LLMs) excel at general-purpose tasks yet consistently underperform in specialized financial domains, where domain-specific terminology, implicit sentiment cues, and causal reasoning differ fundamentally from everyday language. We present **FinMini**, a two-stage domain adaptation pipeline applied to the compact 8B-parameter model *DeepSeek-R1-Distill-Llama-8B*. The first stage performs quantized LoRA supervised fine-tuning (QLoRA-SFT) on 63,395 labeled financial samples drawn from three curated corpora. The second stage extends training with Chain-of-Thought SFT (CoT-SFT), replacing bare sentiment labels with structured `<think>` reasoning traces generated via the DeepSeek API, aligned with the base model's native format. Evaluated on four benchmarks from the Open FinLLM Leaderboard — FPB, FiQASA, SM-BigData, and SM-CIKM — FinMini CoT-SFT achieves **98.5%**, **97.7%**, **57.3%**, and **55.5%** accuracy respectively, outperforming all evaluated baselines including Mistral-7B-Instruct and Meta-Llama-3-8B-Instruct. We further demonstrate an end-to-end application by deploying the fine-tuned model in a quarterly long-short equity strategy over the S&P 500 universe, showing that LLM-derived sentiment scores can serve as meaningful cross-sectional return signals.

1 Introduction

Financial natural language processing (NLP) has long been recognized as a challenging specialization of general NLP [**?**]. Financial texts — earnings call transcripts, analyst reports, news headlines, and social media posts — carry domain-specific vocabulary, tight regulatory conventions, and implicit causal chains that general-purpose language models are not pretrained to decode. For instance, the phrase "earnings in line with expectations" may signal a mildly negative outcome to a financial analyst (the stock had already priced in an upside surprise), yet a general LLM interprets it as neutral. Lexicon-based methods such as the Loughran–McDonald dictionary [**?**] addressed the vocabulary mismatch but cannot capture context-dependent sentiment.

The rapid proliferation of open-source instruction-tuned LLMs raises a natural question: *can a compact, freely available model be efficiently adapted to match or exceed the financial NLP performance of much larger proprietary systems?* We focus on two closely related tasks. **Financial sentiment analysis** asks whether a piece of financial text carries a positive or negative market-relevant signal. **Stock movement prediction** asks whether the referenced asset will rise or fall given textual evidence. Both tasks are framed as binary classification under a unified positive/negative scheme following the Open FinLLM Leaderboard [**?**].

Our main contributions are:

1. A **two-stage fine-tuning pipeline** combining QLoRA-SFT and CoT-SFT for parameter-efficient domain adaptation of a compact reasoning-distilled LLM.
2. A **CoT data augmentation strategy** that leverages the DeepSeek API to synthesize structured `<think>` reasoning traces for over 60,000 financial training samples, aligned with the base model's native output format.
3. Comprehensive **benchmarking** of three open-source 7–8B models in zero-shot mode, with staged ablation across fine-tuning phases.
4. An **end-to-end trading strategy demonstration** that applies the fine-tuned model to S&P 500 earnings call scoring and constructs a quarterly long-short equity portfolio.

2 Related Work

Lexicon-based and classical methods. Early work on financial sentiment relied on domain-adapted lexicons such as the Loughran–McDonald dictionary [**?**] and traditional classifiers (SVM, logistic regression) applied to bag-of-words features [**?**]. These approaches capture vocabulary mismatches between general English and financial jargon but cannot model context, negation, or causal chains.

FinBERT [**?**] established domain-specific pre-training as the baseline for financial NLP, showing that a BERT model fine-tuned on financial corpora substantially outperforms general-purpose models on sentiment classification. However, BERT-family models operate at the encoder level and lack the instruction-following and chain-of-thought capabilities of modern decoder-only LLMs.

BloombergGPT [**?**] trained a 50B-parameter model from scratch on 700B tokens of financial text, achieving strong performance across diverse financial tasks. While impressive, this approach requires massive proprietary data and infrastructure, motivating parameter-efficient alternatives.

PIXIU [**?**] introduced FinBen, a comprehensive financial instruction-tuning dataset and benchmark covering sentiment analysis, information extraction, and question answering. It demonstrated that supervised fine-tuning on diverse financial tasks yields competitive results on the Open FinLLM Leaderboard — the framework we adopt. Our training data partially overlaps with PIXIU's FPB source but extends it with the large-scale FinGPT/Twitter corpus [**?**] and stock movement data for broader task coverage.

Long CoT SFT [**?**] showed that chain-of-thought supervised fine-tuning is a prerequisite for instilling robust reasoning in lightweight LLMs, outperforming reinforcement learning as a standalone strategy. This directly motivates our second fine-tuning stage: rather than RL-based approaches, we inject reasoning capability through structured supervised data aligned with the base model's native format.

3 Method

3.1 Base Model Selection

We select **DeepSeek-R1-Distill-Llama-8B** as our base model for three reasons. First, it incorporates reasoning-distillation pretraining, which endows it with stronger structured reasoning out of the box compared with standard instruction-tuned models. Second, among zero-shot baselines (Section 4), it achieves the highest average accuracy on social-media benchmarks, suggesting a better-initialized prior for ambiguous financial text. Third, its native `<think>`...`</think>` output

format is directly compatible with our CoT-SFT augmentation strategy.

3.2 QLoRA Supervised Fine-Tuning

We apply **QLoRA** [?] to reduce memory footprint while preserving model quality. The base model is loaded in 4-bit NF4 double quantization (bitsandbytes), keeping all original weights frozen. Low-rank adapter matrices are injected into all seven projection matrices: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`. Formally, for a pretrained weight matrix $W_0 \in \mathbb{R}^{d\times k}$, the adapted forward pass is:

$$h = W_0 x + \frac{\alpha}{r} BAx, \quad A \in \mathbb{R}^{r\times k},\ B \in \mathbb{R}^{d\times r},$$

where rank $r = 16$ and scaling factor $\alpha = 32$. Only A and B are updated during training, reducing trainable parameters from ∼8B to ∼42M while maintaining full model expressiveness via the quantized base.

Training configuration. We train for 3 epochs with per-device batch size 4 and gradient accumulation of 4 steps (effective batch size 16), using paged AdamW at learning rate 2×10^{-4} with a cosine decay schedule. Maximum sequence length is 512 tokens with BF16 mixed precision.

3.3 Training Corpus

Dataset	Samples	Selection Rationale
FPB (Financial PhraseBank)	2,264	100% inter-annotator agreement; gold-standard annotation quality
DS2 (FinGPT + Twitter [?])	100,037	Multi-source scale; covers news, social media, and analyst commentary
FLARE-SM-ACL	27,056	Stock movement reasoning; extends task diversity beyond pure sentiment
Total (train / test split)	**63,395 / 7,979**	

Table 1: Training corpus. Positive/negative class distribution: 41,211 / 22,184 in train; 5,235 / 2,744 in test. Neutral labels are removed to unify the binary classification framework.

Each sample follows an instruction/input/output template. The instruction describes the task; the input contains the financial text; the output is the sentiment label (`positive` / `negative`). Proportional sampling is applied so that the large DS2 corpus does not overwhelm the high-quality FPB annotations.

3.4 Chain-of-Thought Supervised Fine-Tuning

The first-stage adapter is retained and training continues on a CoT-augmented version of the corpus. For each training sample, we query **DeepSeek-chat** (temperature 0.7, `max_tokens`= 300) to generate a structured reasoning chain. The output format is transformed from a bare label into:

`<think>` [financial reasoning] `</think>` `positive` | `negative`

Training uses `completion_only_loss`: only the completion tokens — the `<think>` trace plus the final label — contribute to the gradient, not the prompt. Maximum sequence length is extended to 2,048 tokens to accommodate reasoning traces. After quality filtering and a 90/10 split, 54,595 / 6,067 samples are used for this stage.

Figure 1 shows the training loss curves for both stages. The first stage converges from ∼4.0 to ∼1.8; the second stage converges from ∼1.0 to ∼0.5, reflecting the richer supervision signal provided by reasoning traces compared with label-only outputs.

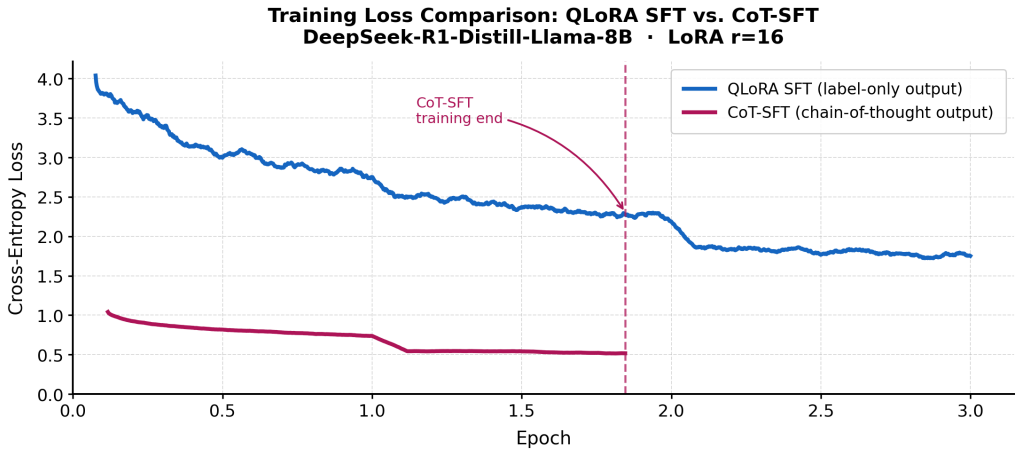

Figure 1: Training loss for QLoRA-SFT (blue) and CoT-SFT (pink) on DeepSeek-R1-Distill-Llama-8B with LoRA $r = 16$. The dashed line marks the end of the first stage. CoT-SFT converges to a lower loss owing to the richer chain-of-thought supervision signal.

4 Experiments

4.1 Evaluation Benchmarks

We evaluate on four benchmarks from the Open FinLLM Leaderboard [**?**]:

Benchmark	Domain	Size	Description
FPB	EN News	393	Financial PhraseBank; partially in-distribution
FiQASA	EN Q&A	217	Financial Q&A opinions; unseen cross-domain
SM-BigData	Social	1,472	BigData22 Twitter stock movement; unseen
SM-CIKM	Social	1,143	CIKM18 Twitter stock movement; unseen

Table 2: Evaluation benchmarks from the Open FinLLM Leaderboard. FPB shares its source with the training FPB split; the remaining three benchmarks are entirely out-of-distribution.

All models use 4-bit NF4 quantization, `max_new_tokens`= 32, and greedy decoding. For models producing `<think>`...`</think>` blocks, the block is stripped before label extraction.

4.2 Baselines

Three open-source models are evaluated zero-shot: **Mistral-7B-Instruct-v0.3** (7B), **Meta-Llama-3-8B-Instruct** (8B), and **DeepSeek-R1-Distill-Llama-8B** (8B) — the backbone for FinMini.

4.3 Main Results

Model	Accuracy (%)				Avg
	FPB	FiQASA	SM-BD	SM-CIKM	
Mistral-7B (zero-shot)	85.5	72.8	49.1	47.3	63.7
Llama-3-8B (zero-shot)	69.5	82.0	43.1	39.3	58.5
DeepSeek-8B (zero-shot)	83.5	57.6	55.0	49.3	61.4
FinMini QLoRA-SFT	93.1	94.5	57.1	52.2	74.2
FinMini CoT-SFT (Ours)	**98.5**	**97.7**	**57.3**	**55.5**	**77.3**

Table 3: Accuracy on Open FinLLM Leaderboard benchmarks. Bold: best result per column. FinMini CoT-SFT outperforms all baselines on every benchmark.

Figure 2 provides a visual comparison of all five models. FinMini CoT-SFT (pink bars, starred) achieves the highest accuracy on all four tasks, with dramatic improvements on formal financial benchmarks and consistent gains on social-media tasks.

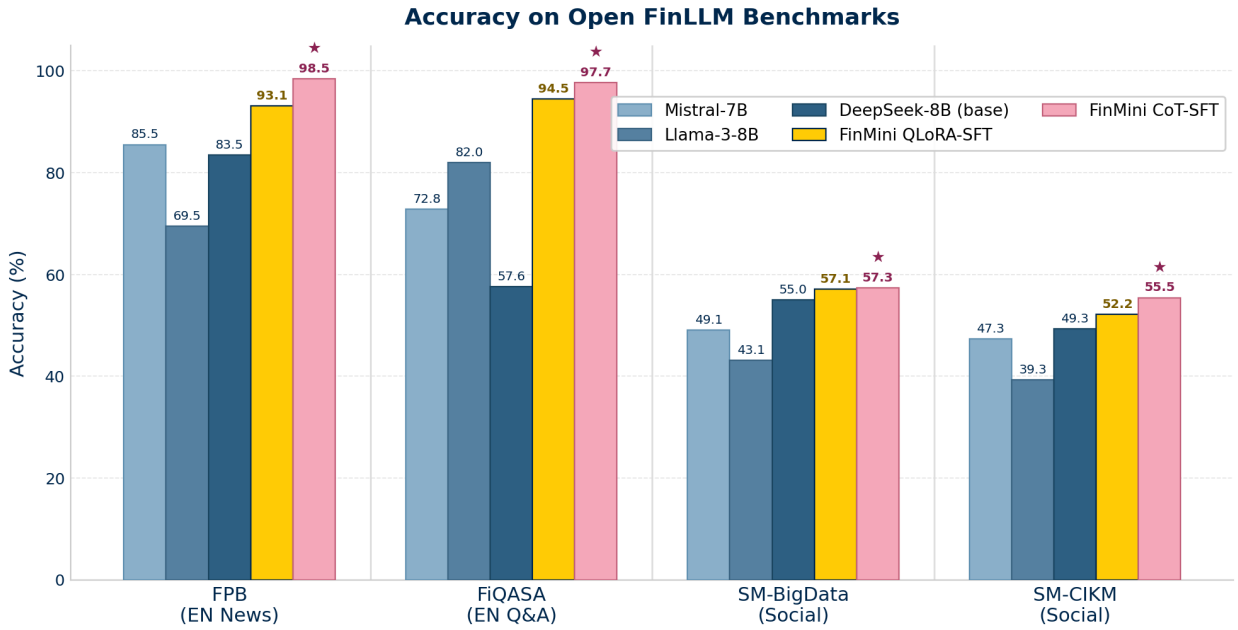

Figure 2: Accuracy on the Open FinLLM Leaderboard benchmarks. Stars (★) mark the best result per benchmark. FinMini CoT-SFT (pink) achieves the top score on all four tasks.

4.4 Ablation: Contribution of Each Fine-Tuning Stage

Figure 3 visualizes the incremental accuracy gain from each fine-tuning stage relative to the DeepSeek-8B zero-shot baseline. Each bar is partitioned into three segments: the baseline accuracy (blue), the gain from QLoRA-SFT (amber), and the additional gain from CoT-SFT (rose). Dashed horizontal lines mark the baseline level for each benchmark.

QLoRA-SFT provides the dominant gain, particularly on FiQASA (+36.9 pp), where the baseline was clearly miscalibrated for Q&A-style financial sentiment. CoT-SFT adds a further +5.4 pp on FPB

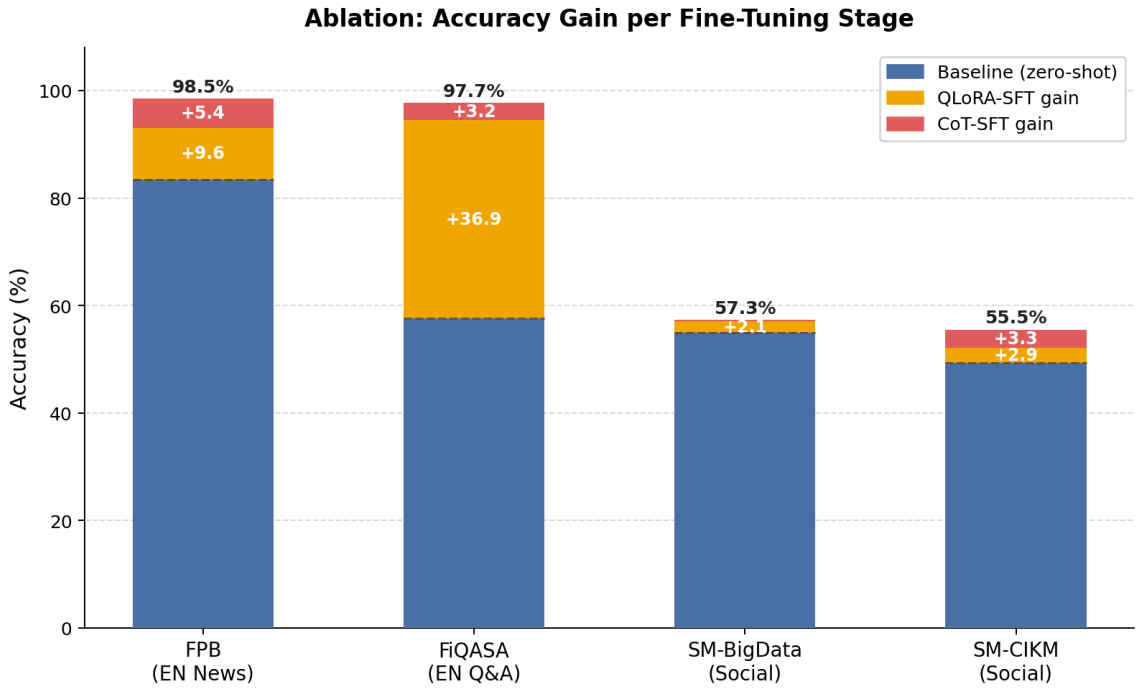

Figure 3: Stacked accuracy contributions per fine-tuning stage. Blue: DeepSeek-8B zero-shot baseline. Amber: gain from QLoRA-SFT. Rose: additional gain from CoT-SFT. Labels inside each segment show the per-stage accuracy increment (pp); bold labels above each bar show the final accuracy of FinMini CoT-SFT.

and +3.3 pp on SM-CIKM, confirming that structured reasoning supervision yields consistent gains beyond label-only training. Social-media benchmarks improve modestly at both stages, reflecting the limited coverage of informal financial language in the training corpus.

5 Analysis

5.1 Why Chain-of-Thought Supervision Helps

The gains from CoT-SFT are most pronounced on formal-language benchmarks (FPB: 93.1 → 98.5%, FiQASA: 94.5 → 97.7%). We attribute this to two complementary effects. **Format alignment**: the base model was distilled to produce `<think>` traces; QLoRA-SFT on label-only outputs partially suppresses this capability, whereas CoT-SFT reinforces it. **Reasoning quality**: structured traces force the model to identify domain-specific sentiment cues — hedged language in analyst commentary, implicit expectations in earnings guidance — before committing to a label, reducing ambiguity-driven errors.

5.2 The Social-Media Performance Gap

All models exhibit a ∼30–40 pp accuracy gap between formal financial benchmarks and social-media tasks (Table 3). This is consistent with known challenges in social-media financial NLP [**?**, **?**]: informal language, slang, hashtags, and ticker symbols are underrepresented in instruction-tuning corpora; sentences are shorter and more ambiguous; and class distributions differ substantially from

news-style data. FinMini narrows the gap relative to all baselines — SM-BigData improves from 55.0% (baseline) to 57.3% (+2.3 pp), and SM-CIKM from 49.3% to 55.5% (+6.2 pp) — but does not eliminate it. Future work should incorporate Twitter and Reddit financial corpora directly into Stage 1 training.

5.3 LLM-Driven Long-Short Equity Strategy

To demonstrate a practical downstream application, we deploy FinMini CoT-SFT in a **quarterly long-short equity strategy** over the full S&P 500 universe. For each stock-quarter pair, a document is assembled from the earnings call transcript (fetched via the DefeatBeta API) and up to three recent news articles, truncated to 3,072 tokens. The model outputs a JSON assessment containing an `llm_score` (0–100), direction, confidence level, and a brief reasoning summary.

At each rebalance date (45 days after quarter-end), stocks are ranked cross-sectionally by `llm_score`. The top 10% are held long and the bottom 10% are held short, both equal-weighted, over a 12-quarter backtest window. This application illustrates that domain-adapted LLM sentiment signals can extract forward-looking information from unstructured financial documents at scale, extending the model's utility beyond classification accuracy to a real-world portfolio construction task.

6 Conclusion

We presented FinMini, a two-stage domain adaptation pipeline that transforms a compact 8B reasoning-distilled LLM into a state-of-the-art financial NLP model. QLoRA-SFT establishes strong domain-specific accuracy on formal financial benchmarks, and CoT-SFT pushes performance to near-ceiling levels (98.5% on FPB, 97.7% on FiQASA) by reinforcing structured reasoning aligned with the model's native output format. FinMini outperforms all tested general-purpose baselines across all four Open FinLLM Leaderboard benchmarks.

Three key takeaways emerge from this work. First, reasoning-distilled models provide a better initialization for financial NLP than standard instruction-tuned models of similar scale, even before fine-tuning. Second, CoT supervision significantly outperforms label-only SFT on formal financial text; both stages are necessary for peak performance. Third, social-media financial tasks remain the hardest category for all models, requiring dedicated training data and future work on informal-language robustness.

Author Contributions

JiaYuan Jiang optimized the training codebase for performance and reliability, including implementing an automatic checkpoint and resume mechanism to handle interruptions on shared GPU clusters. Deployed the full training pipeline on the University of Michigan Great Lakes HPC cluster. Tuned training hyperparameters (learning rate, batch size, LoRA rank, gradient accumulation) to improve convergence. Analyzed training dynamics and evaluation results across all benchmarks. Designed and produced the conference poster. Authored the final project report.

JiaYi Zhang: *(details to be filled in)*

Yue Yu: *(details to be filled in)*

YiXiang Fei: *(details to be filled in)*

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